Feature F11 Innovation Rationale

Wind Tunnel synthetic screening rehearsal

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1 Problem and purpose

Recruitment screening can produce scores even when a job description contains ambiguous requirements, compound criteria or unsuitable weights. Recruiters discover these problems after reviewing real applications, making corrections harder to explain and compare. Conventional CV scoring evaluates applicants but provides limited visibility into how the screening configuration itself behaves.

Feature F11, the Wind Tunnel, addresses this problem through synthetic screening rehearsal. It helps recruiters inspect a saved job configuration before relying on its results. The purpose is to expose possible interpretation issues and support deliberate improvements while preserving employer intent and human decision authority.

2 Architecture and workflow

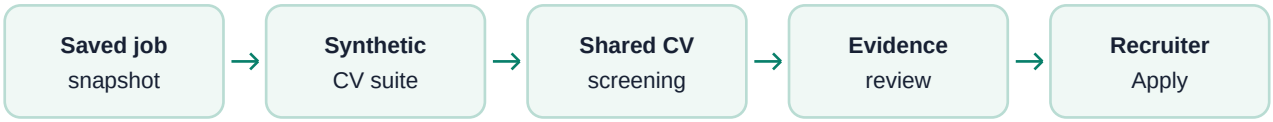
An eligible job approval creates a durable task; recruiters can request suggestion batches. A background worker claims the task and executes a LangGraph workflow using a versioned snapshot of the job description, screening profile and processing configuration.

The current suite generates two CV personas: strong_all_rounder and keyword_stuffer. Each receives one assessment pass. Validated documents pass through extraction, parsing, evidence matching and deterministic scoring using shared CV processing components. Synthetic documents, assessments and retrieval resources remain separate from real applications.

The workflow evaluates observed results and fixed-match weight sensitivity, then generates grounded job-description or criteria proposals. Validated proposals appear in the recruiter workspace. The recruiter previews selected changes together and explicitly confirms Apply; generation never directly changes or publishes the job.

FastAPI exposes authenticated job-level suggestion endpoints, while the React workspace presents batch progress, saved findings and proposal previews. PostgreSQL stores task state and diagnostic artifacts; document storage and scoped retrieval support source inspection.

F11 WORKFLOW



Technical Design and Value

Implementation rationale and operating boundaries

3 Design choices

Reusing the existing screening pipeline avoids a parallel scoring implementation and makes rehearsal behavior relevant to ordinary screening. Separate synthetic records prevent test profiles from contaminating candidate lists, interview invitations or hiring reports.

Persisted snapshots, source hashes and processing versions make findings traceable to specific inputs. Durable tasks, leases and saved progress support eligible recovery after interruptions. Source-version checks reject stale proposals when the job changes.

The application validates proposal structure, supporting evidence and combined edits. Model-based semantic review adds another check, but does not replace deterministic validation or recruiter judgment. Ownership checks, review tokens and explicit confirmation protect the final mutation. Sensitivity analysis transfers five percentage points between dimensions while holding matches fixed.

Applying proposals reuses the established job editor, preserving one authoritative mutation path. The reviewed description and criteria are saved together, and other ready suggestions become stale after the job changes.

4 Practical value

F11 gives recruiters a way to question screening behavior instead of accepting numerical output without context. It supports clearer requirements, inspectable scoring assumptions and traceable editorial decisions. For the project, its innovation is evaluating the screening configuration itself, complementing features that evaluate candidates. The architecture also demonstrates reusable workflows, isolated test data, controlled AI actions and recoverable background processing.

A recruiter can examine how an evidence-rich profile differs from a keyword-heavy profile, then inspect the saved criterion-level explanations before deciding whether a proposed clarification is appropriate.

5 Operating boundary

F11 is a synthetic diagnostic aid. Its small rehearsal suite does not establish statistical confidence, and proposed changes require recruiter review before application.

Implementation references

backend/app/workflows/wind_tunnel/graph.py; services/wind_tunnel/policy.py, evaluation.py and review.py; models/wind_tunnel.py.

Paths after the first reference are relative to backend/app. Current implementation reviewed; no benchmark or fresh test pass is asserted.